MANAGEMENT SYSTEM

Developer Integration & Test Blueprint

System interfaces • Permission policy • Workflow gates • Delivery acceptance

Plan better, build better.

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Status: Business requirements harmonized • Developer implementation not yet performed

01. Developer scope and working agreements

Build one secure, mobile-capable management platform with seven domain modules plus common portals. Distinct project modules are joined by canonical project identifiers, signed approvals, event history and independent role permissions. Module 7 supplies common services continuously; do not build it as a project step that starts only after construction.

- Deliver responsive employee (including low-bandwidth skilled worker), supervisor, PM, department, GM, protected auditor, owner, subcontractor, supplier and client role experiences.
- Use strong authentication/MFA for approvals/payee changes and protected audit; role-based plus project-scoped authorization validated on server.
- Never silently edit original submitted evidence. Expose corrected/reviewed versions and preserve immutable original to authorized auditors.
- Treat permission, contract, statutory, financial and safety gates as server-side invariants, not only disabled browser buttons.
- Provide API docs, tests, migrations, backups, staging deployments, source ownership handover and staff training.

02. Top-level navigation

Portal	Minimum screens
Intake / Site Visits	Omnichannel lead inbox, agreed booking wizard, site survey mobile capture, generated report, lead review, publication.
Design	Department pipeline boards, task assignment, 2D client review, 3D, working-file revisions, engineering discipline QA, QS budget.
Client Proposal	Summary/detailed finance controls, published revisions, identity/acceptance, receipts, exception and handover to M4.
Approvals	Authority case queue, regulatory obligation checklist, controlled issue sets, fee proofs, correction tasks, conditional close.
Contracts	Service selection, agreements and versions, negotiation register, management signed offer, acceptance and signatures, PM appointment.
Construction	PM overview, supervisor tasks, worker personal portal, original report comparison, GRN, inventory, supplier, finance forecast, equipment, safety, claims and warranty.
Administration	HR, staff records and privileges; central Finance/Accounts; supplier master; QA/Compliance; auditor; GM/company administration.
Client	Projects, signed 2D, approved PDFs/images, budget acceptance, billing, final contracts, PM updates, warranties.
Owner / executive	Owner strategic governance and independent oversight; GM operational command separate.
Settings	Authority matrix, thresholds, roles, credential types, licenses, statutory checklists, integrations, retention, backups.

03. Proposed backend service boundaries

Domain / API example	Authorization and business invariant
POST /enquiries/{id}/confirm-visit	Agreement evidence required to instantiate ProjectId.
POST /projects/{id}/surveys	Project membership; creator original cannot be overwritten.
POST /projects/{id}/design/2d-acceptances	Valid exact 2D revision and client identity/acceptance proof.
POST /design-packages/{id}/technical-approvals	Authorized qualified discipline lead; predecessor stage

Domain / API example	Authorization and business invariant
	satisfied.
POST /projects/{id}/proposals/{id}/publish	Eligible manager grant and QS approved exact budget version.
POST /projects/{id}/client-acceptances	Client or trusted uploaded signed evidence, immutable revision.
POST /projects/{id}/module4-releases	Management acceptance checked and Accounts verified milestone/exception.
POST /projects/{id}/compliance/cases	Applicable jurisdiction obligation and authorization.
POST /contracts/{id}/execute	Final terms approved; signer legal mandate; counterpart evidence.
POST /projects/{id}/mobilization	Accounts gate, compliance permit, PM readiness, delegated physical-start decision.
POST /projects/{id}/worker-reports	Worker-only own report; immediate independent manager visibility.
POST /worker-reports/{id}/verifications	Separate supervisor review; originals immutable.
POST /purchase-orders/{id}/amendments	Version-specific QS/Accounts reviews, management authorization, supplier notice.
POST /goods-receipts	Storekeeper physical actual accepted quantity, batch, photos and PM verification.
POST /suppliers/{id}/beneficiary-changes	Out-of-band known-contact verification and dual-control; payment hold.
POST /assets/{id}/pre-use-inspections	Equipment-licensed operator check; serious failure creates targeted hold.
POST /fuel-issues	Measured liters/quantity, asset, operator/site/custody, stores reconciliation.
POST /assets/{id}/return-to-service	Qualified independent release; no repairer self-approval.
POST /projects/{id}/variation-acceptances	Exact version, client accepted, manager approved, QS and required technical approval.
POST /projects/{id}/handover	PM technical pack, accounts/contract check, authorized management release.
POST /audits/cases	Protected auditor workflow; no manager edit to independent originals.

04. Common request and evidence schema

Suggested minimal request envelope; naming and exact object shape are developer choices, while the semantics below are mandatory.

Field	Type / rule
id / projectId / engagementId	Stable identifiers; project references may not change between modules.
entityType / entityId / action	Controlled resource, action and audit intent.
actorId / representedRole	Actual authenticated identity, acting authority and role at time of action.
authorityGrantId	Specific valid mandate for actions requiring delegated approval.
timestamp / clientSubmittedAt	Authoritative server timestamp plus claimed device time when offline.
sourceRevision / latestApprovedRevision	Hash/version protection; old client signatures still point to old version.
originalEvidenceHashes / attachmentRefs	Evidence ownership, object authorization, checksum and retention.
previousState / nextState / reason	Append-only workflow transition; exception reason and approver.
idempotencyKey / correlationId	Prevent duplicate writes and trace event flows.
visibilityClass / publicationGrantId	Role/project-specific confidentiality and exact client file

Field	Type / rule
	publication.

05. End-to-end integration sequence

- Enquiry -> confirmed visit -> project ID -> mobile original survey -> generated site report -> architect review/assign -> Module 2.
- Architectural 2D -> client versioned approval -> 3D -> structural -> mechanical/plumbing -> electrical -> QS -> proposal snapshot.
- Management-approved publication -> client portal/physical approval -> management verifies -> Accounts checks milestone -> separate Module 4 release.
- Applications and professional certification -> corrections/version control -> conditional planning approval -> contracted service choice and sign.
- Executed contract -> PM appointed -> planning-only construction -> Accounts and Compliance readiness -> manager physical mobilization authorization.
- Worker tasks and originals -> supervisor verification -> PM analysis and authorized client publication; store and asset events link project cost.
- Procurement approved PO -> supplier -> actual accepted GRN -> match invoicing -> independent authorized payment and Accounts reconciliation.
- Safety / quality / statutory hold -> independent qualified review -> documented clearance -> authorized affected work release.
- Variations / milestone billing / time impacts -> QS/Contracts/Accounts -> management approved client terms and signature.
- Practical completion -> separate handover, financial final account, retention, defects, warranty case and final closure.

06. Non-negotiable business invariants

- Platform owner must not inherit unrestricted superuser audit data or daily operational approvals just because they own shares.
- GM and departmental managers cannot unilaterally alter independent whistleblower evidence or audit case attribution.
- No automated arrears reminders: Accounts must explicitly initiate and log each client payment reminder.
- No construction use of expired/unsafe asset, unqualified operator, statutory noncompliant site or professionally uncleared structural stage.
- Financial exceptions only affect eligible financial gate and retain actual balance; they never forgive debt or override statute.
- Never issue a purchase order from a request alone. Supplier evaluation, commitments, management authorization and PO remain separate.
- Never pay changed supplier bank/M-Pesa details until independent out-of-band trusted-contact verification and reauthorization.
- No wage deduction as an automatic variance/loss response; all responsibility decisions require fair and lawful procedure.
- Advance accounting impairment differs from legal waiver or debt extinguishment; audit both.
- Opening Module 7 controls does not automatically activate a project stage; service scope per client contract is authoritative.

07. Acceptance tests and expected behavior

ID	Test situation	Expected system result
UAT-01	Booking without client-agreed site date	Reject project activation and record enquiry only.
UAT-02	Junior 3D attempted before 2D client signature	Reject; show required exact 2D version and evidence.
UAT-03	Client accepts proposal then QS revises quantities	Preserve original, require applicable reacceptance, block old acceptance from satisfying new version.
UAT-04	Accounts inserts unverified mobile money receipt	Do not clear milestone or generate “verified” payment receipt.
UAT-05	Automated nightly client debtor reminder schedule	Must not send unless Accounts deliberately initiates each reminder.
UAT-06	Approvals Officer stamps professional drawing	Deny, independent verified professional required.
UAT-07	Conditional Module 4 complete but NCA pending	Construction start blocked; no fake “completed NCA” state.
UAT-08	Client selects labour-only, does not sign	Do not create active binding labour contract.
UAT-09	Owner account tries routine pay authorization	Deny without distinct valid assigned operational role/exception; still allow strategic oversight.
UAT-10	GM away with registered deputy	HR-valid preapproved mandate activates only permitted time-limited powers.
UAT-11	GM absent with no valid deputy	Preapproved emergency committee logs quorum and limited decisions; never owner-reserved.
UAT-12	Worker uploads photographic report and supervisor rejects	Preserve original; PM and GM see it without supervisor gating; rejection is separate.
UAT-13	Supervisor tries to delete accident evidence	Deny, preserve original and audit attempted action.
UAT-14	Supplier changes M-Pesa/bank destination with pending payment	Hold affected payment, trusted-contact verify, and require revised authorization.
UAT-15	Supplier invoice value larger than accepted GRN	Reject noncontracted overpayment, retain partial delivery balance.
UAT-16	PO revision after supplier award	Version, verify independent QS/Accounts and delegated approval before commitments.
UAT-17	Receiving store checks one of two batches only	Only actual independently accepted stock enters receiving store; in-transit remainder distinct.
UAT-18	Client-owned stock transfer to other project	Block without verified title/contract entitlement and required client consent.
UAT-19	Loss reported against worker	Investigate independently; no automatic financial liability or pay deduction.
UAT-20	Operator with expired qualification accesses excavator	Block assignment/operation and alert qualified staff.
UAT-21	Critical pre-use inspection failure on concrete mixer	Targeted asset hold, repair and independent technical release; unaffected assets available.
UAT-22	Fuel diesel issue without machine/project and recipient	Reject record as verified issue until required data/measurement present.
UAT-23	Site Supervisor requests extra chargeable work	Do not execute before professional/QS/Contracts and management-approved client

ID	Test situation	Expected system result
		acceptance.
UAT-24	Accounts or PM tries publishing worker raw dispute to client	Deny without valid publication grant and project-scoped redaction.
UAT-25	GM subject of fraud allegation	Protected audit route bypasses GM and escalates to nonconflicted governance.
UAT-26	Site inspector fails reinforcement hold point	No concreting release until competent approval and PM authorization.
UAT-27	Accounts issues final subcontract retention by expiry alone	Blocked until contract conditions and delegated authorization checked.
UAT-28	Warranty period expires with open client complaint	Keep case and all obligations open for proper determination.
UAT-29	Project marked handed over before financial reconciliation	Track handover and financial close as separate explicit pending/completed statuses.
UAT-30	User guesses cross-project attachment URL	Return forbidden; no file bytes or metadata leaked.

08. Non-functional requirements and evidence

- Supported use: responsive browser for operations; low-bandwidth mobile employee capture with offline queue and explicit sync acknowledgment.
- Availability and recovery targets must be separately approved; implement encrypted offsite backups, restore exercises, encrypted transport, confidentiality and disaster recovery.
- Use queue-based resilient notifications; if WhatsApp/TikTok/Facebook/YouTube integration is not approved/available, fall back to staff task queue instead of pretending messages were sent.
- Create staging test users for Owner, GM, acting GM, HR, Accounts requester/payee verifier/payer/reconciler, QS, Architects, PM, supervisors, workers, subcontractors, auditors and clients.
- Provide documented role matrix, threat model, professional legal regulatory checklist, event-to-record traceability, audit retention and case confidentiality proofs.
- Write automatic tests for every table row above, show evidence screenshots and the underlying exact records in demonstrations, and obtain signed acceptance per delivery wave.

09. Company configuration required before production

Configuration	Who validates / supplies
GM delegation caps, owner-reserved matters, emergency quorum	Owner governance + corporate signatories + General Manager
Staff members, payroll classes, contract types and lawful deductions	HR + Accounts + legal advisers
County, NEMA, NCA, statutory forms and expiry durations	Approvals Officer + qualified professionals + Compliance
Trade/plant licensing, daily checklist and maintenance	Technical/Safety + HR + Project Manager
Procurement quoting thresholds, single-source exception, approved suppliers	Procurement + QS + Accounts + GM
Account codes, taxes, funding, retention and payments	Accounts/Finance + external accounting adviser
Client document publication, portal channels and notification cadence	GM/dept heads + Contracts + client project agreement
Backup period, retention, privacy response and forensic access	IT Security + HR + Audit + legal advisers

10. Handover pack and development completion checklist

- Approved V3 module specifications and master architecture, version register and change-control process.
- Interactive wireframes/user role navigation, mobile-first worker and client flows and accessible field labels.
- Entity relationship diagram and data dictionary, migration scripts, API schema and example state transitions.
- Security model, identity, confidential audit access, split duties and recovery controls.
- Automated integration/unit/UAT suite and sample staging projects for design-only and full construction, including negative tests.
- Runbooks, user/admin training, monitoring, error reporting, source/code ownership and deployment documentation.